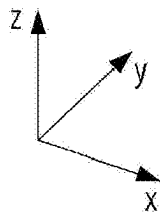
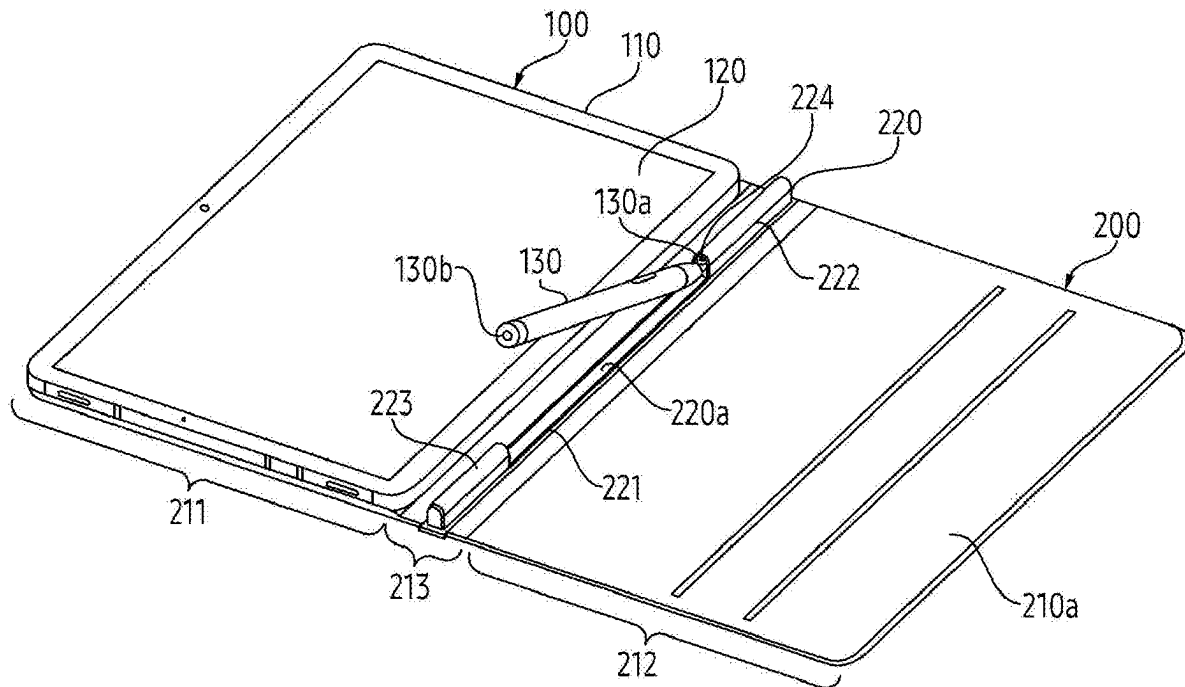




US 20250284314A1

(19) **United States**(12) **Patent Application Publication**
KIM(10) **Pub. No.: US 2025/0284314 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **CASE INCLUDING STRUCTURE FOR
RECEIVING ELECTRONIC PEN**(52) **U.S. Cl.**
CPC **G06F 1/1628** (2013.01); **G06F 3/03545**
(2013.01)(71) Applicant: **Samsung Electronics Co., Ltd.**,
Suwon-si (KR)(72) Inventor: **Taeyang KIM**, Suwon-si (KR)(57) **ABSTRACT**(21) Appl. No.: **19/219,151**(22) Filed: **May 27, 2025****Related U.S. Application Data**(63) Continuation of application No. PCT/KR2023/
017114, filed on Oct. 31, 2023.(30) **Foreign Application Priority Data**Jan. 12, 2023 (KR) 10-2023-0004889
Feb. 6, 2023 (KR) 10-2023-0015506**Publication Classification**(51) **Int. Cl.**
G06F 1/16 (2006.01)
G06F 3/0354 (2013.01)

A case detachably coupled to an electronic device is provided. The case includes a plate supporting the electronic device and a storing member for storing at least a portion of an electronic pen disposed on a surface of the plate, wherein the storing member includes a receiving space receiving an end of the electronic pen, a first actuating portion applying a force in a direction toward the receiving space to another end of the electronic pen opposite to the end of the electronic pen such that the end of the electronic pen is inserted into the receiving space, and a second actuating portion, disposed between the receiving space and the first actuating portion, and applying a force in a direction toward the surface of the plate to the electronic pen such that the electronic pen is seated onto the storing member, and wherein a portion of the storing member is open toward an outside of the storing member such that at least a portion of the electronic pen is exposed outside the storing member.

210 { 211
212
213

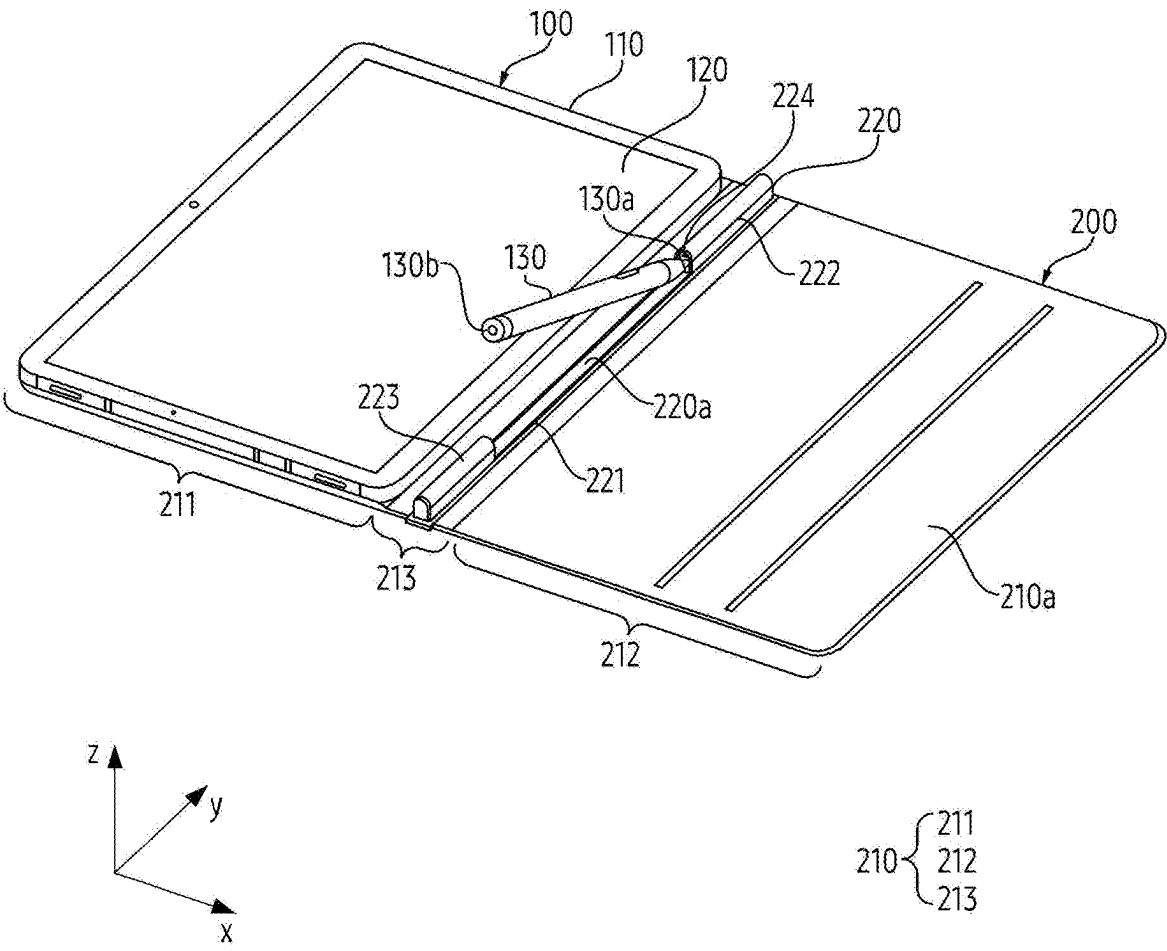


FIG. 1A

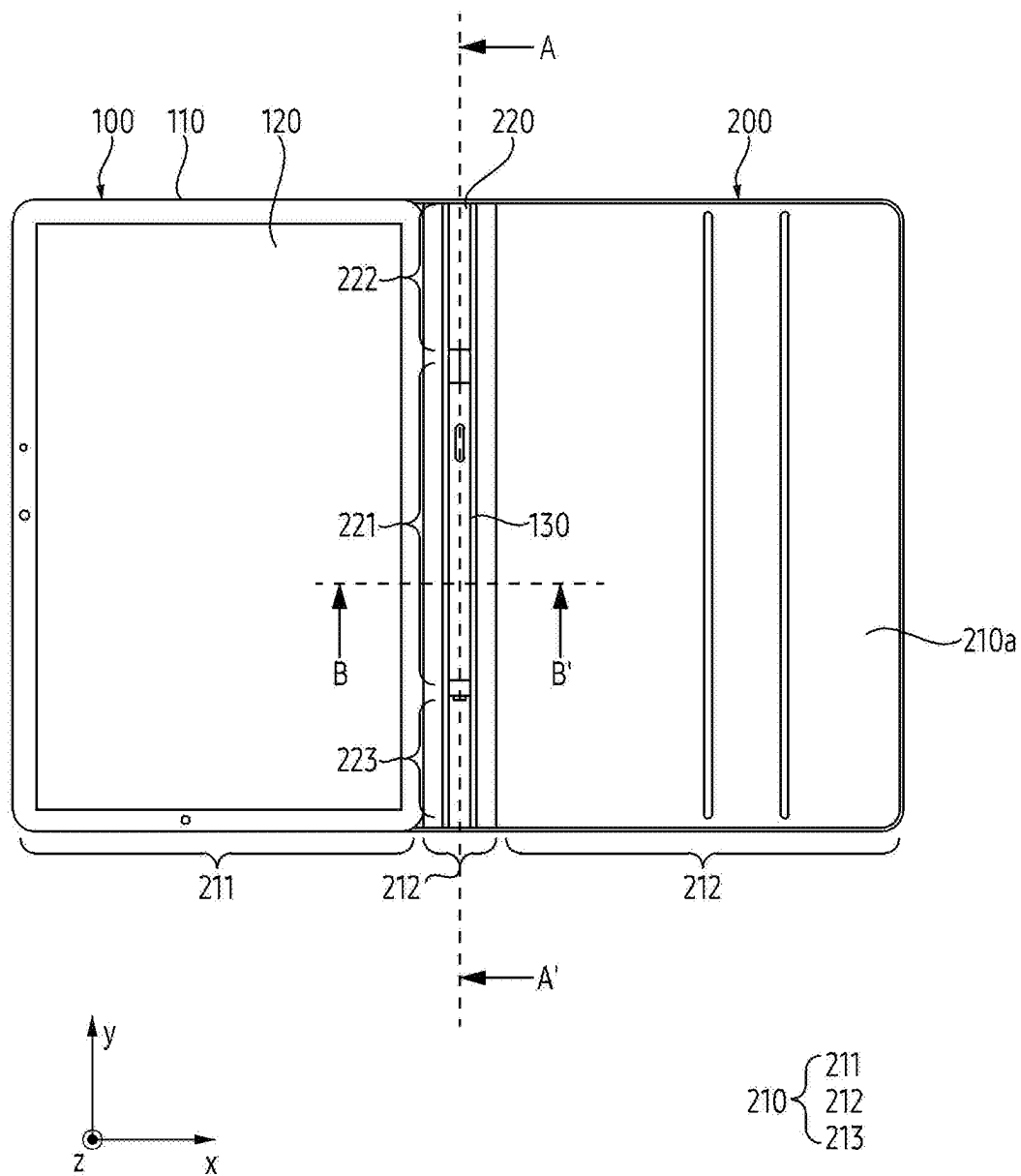


FIG. 1B

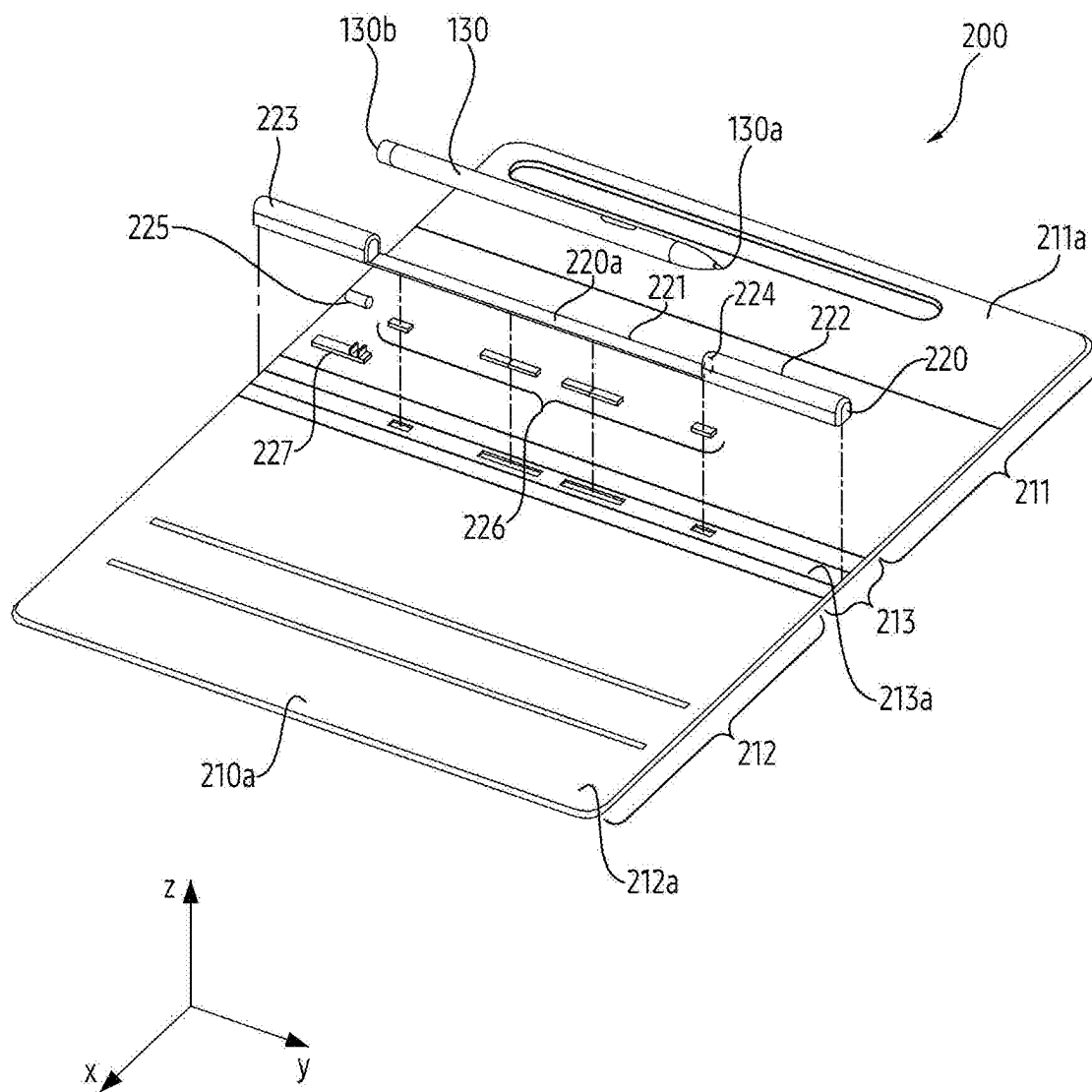


FIG. 1C

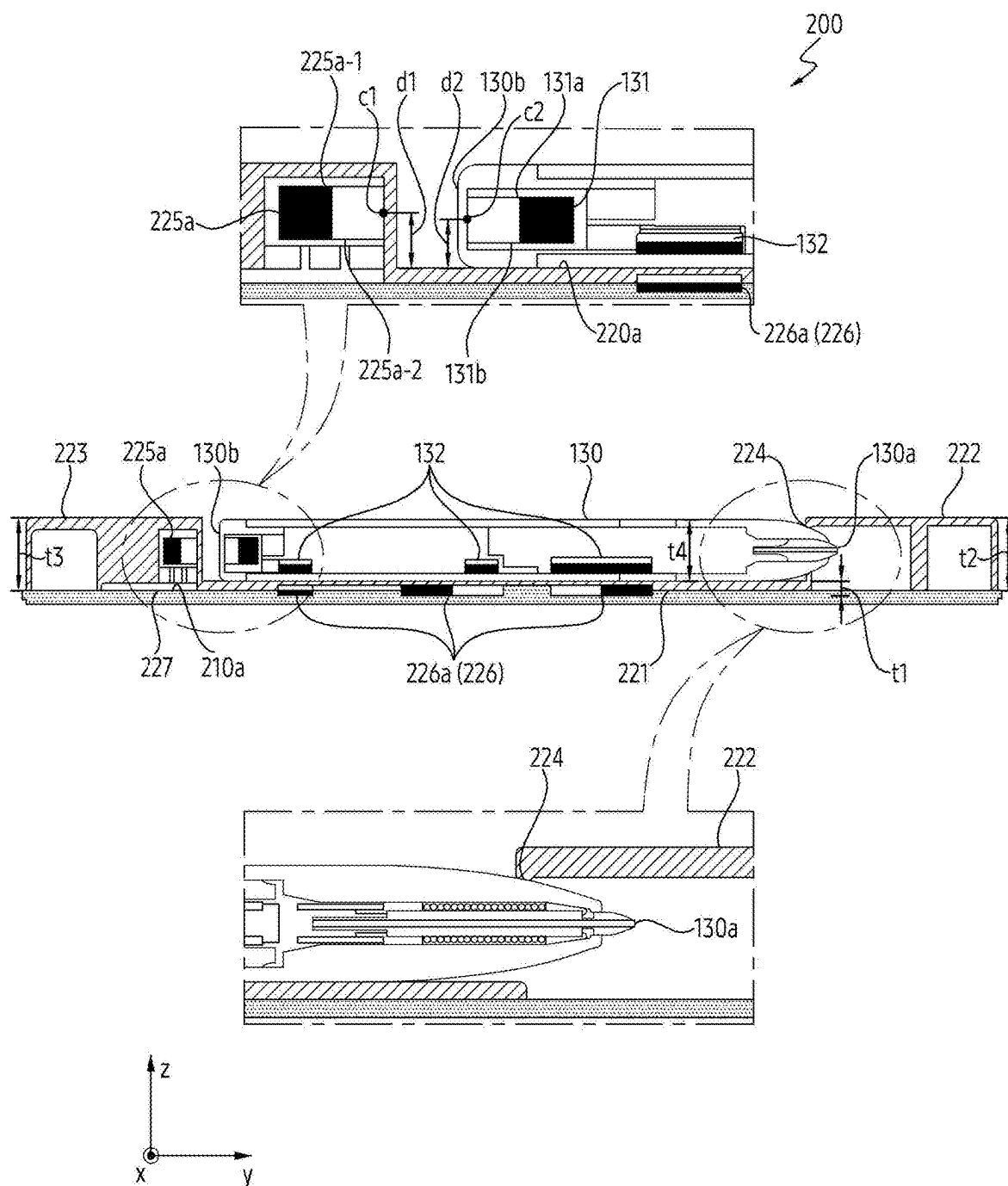


FIG. 2

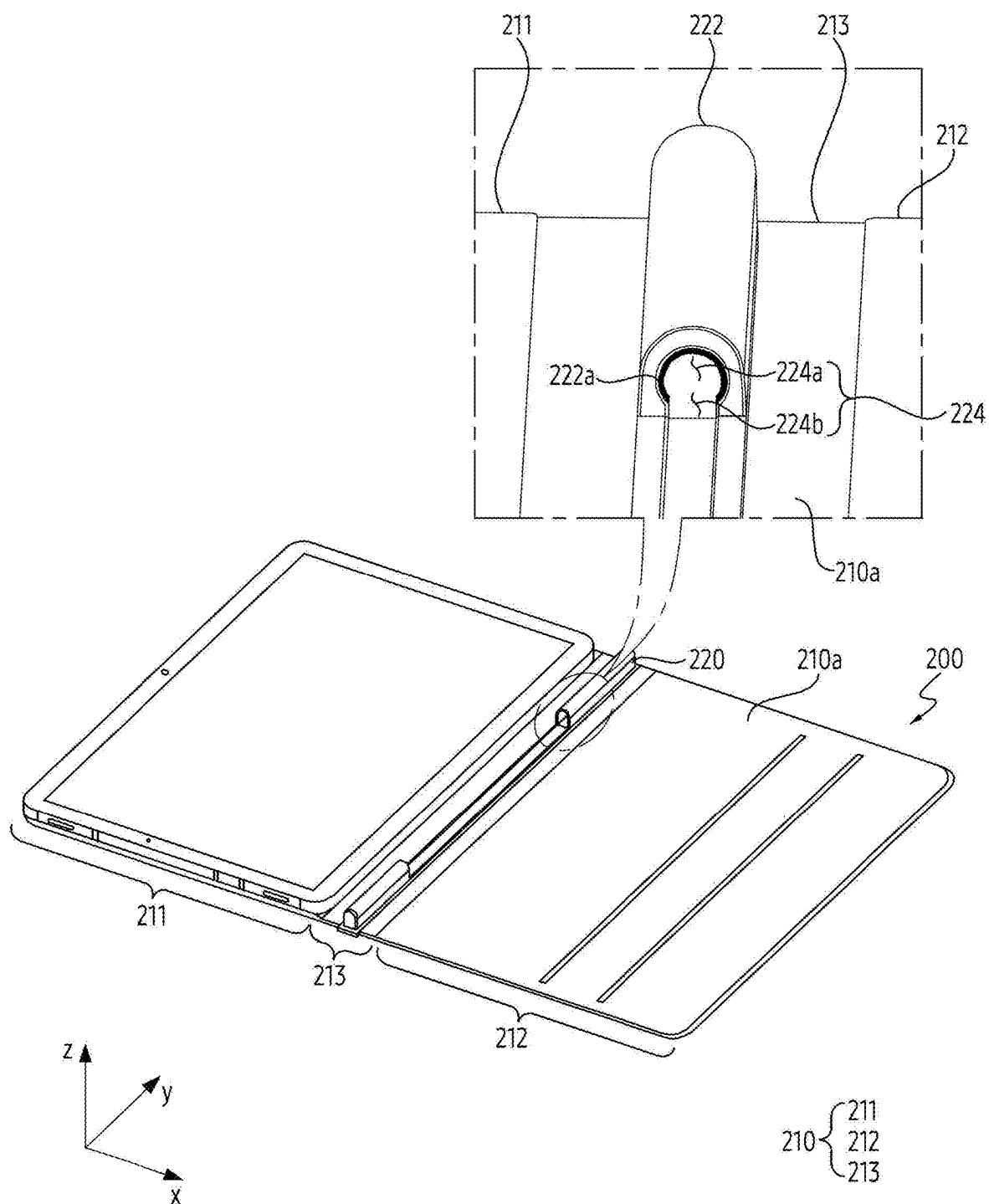


FIG. 3A

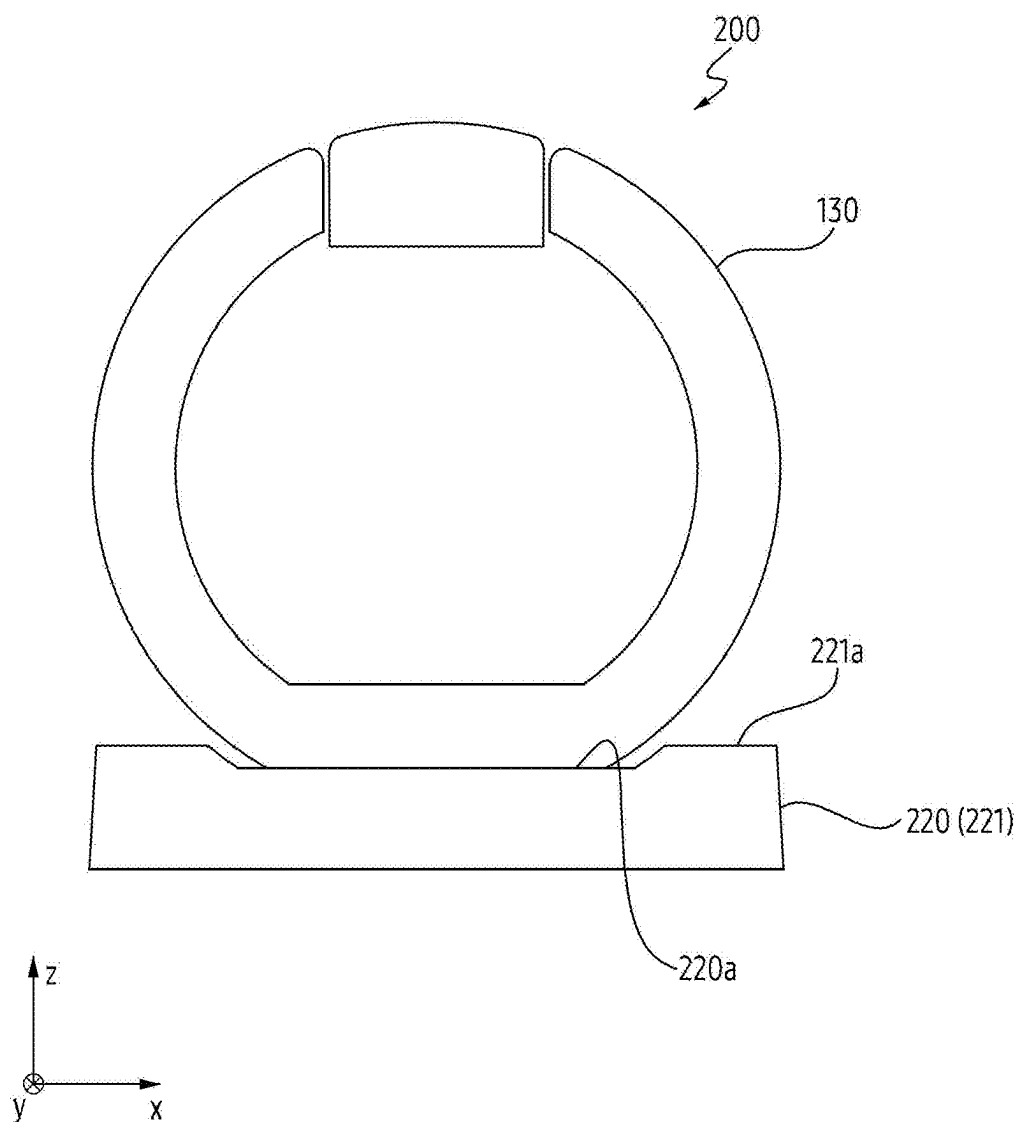


FIG. 3B

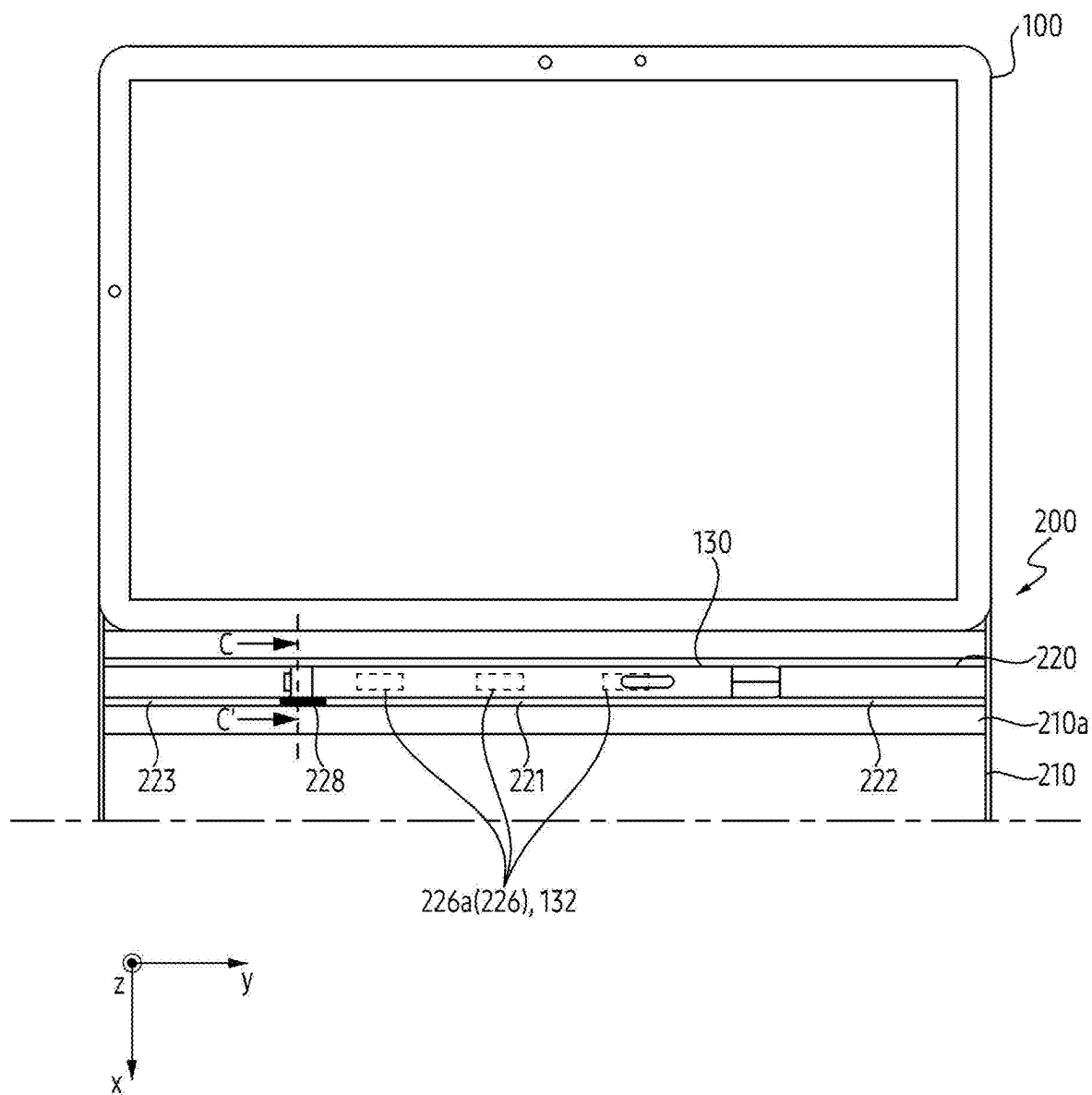


FIG. 4A

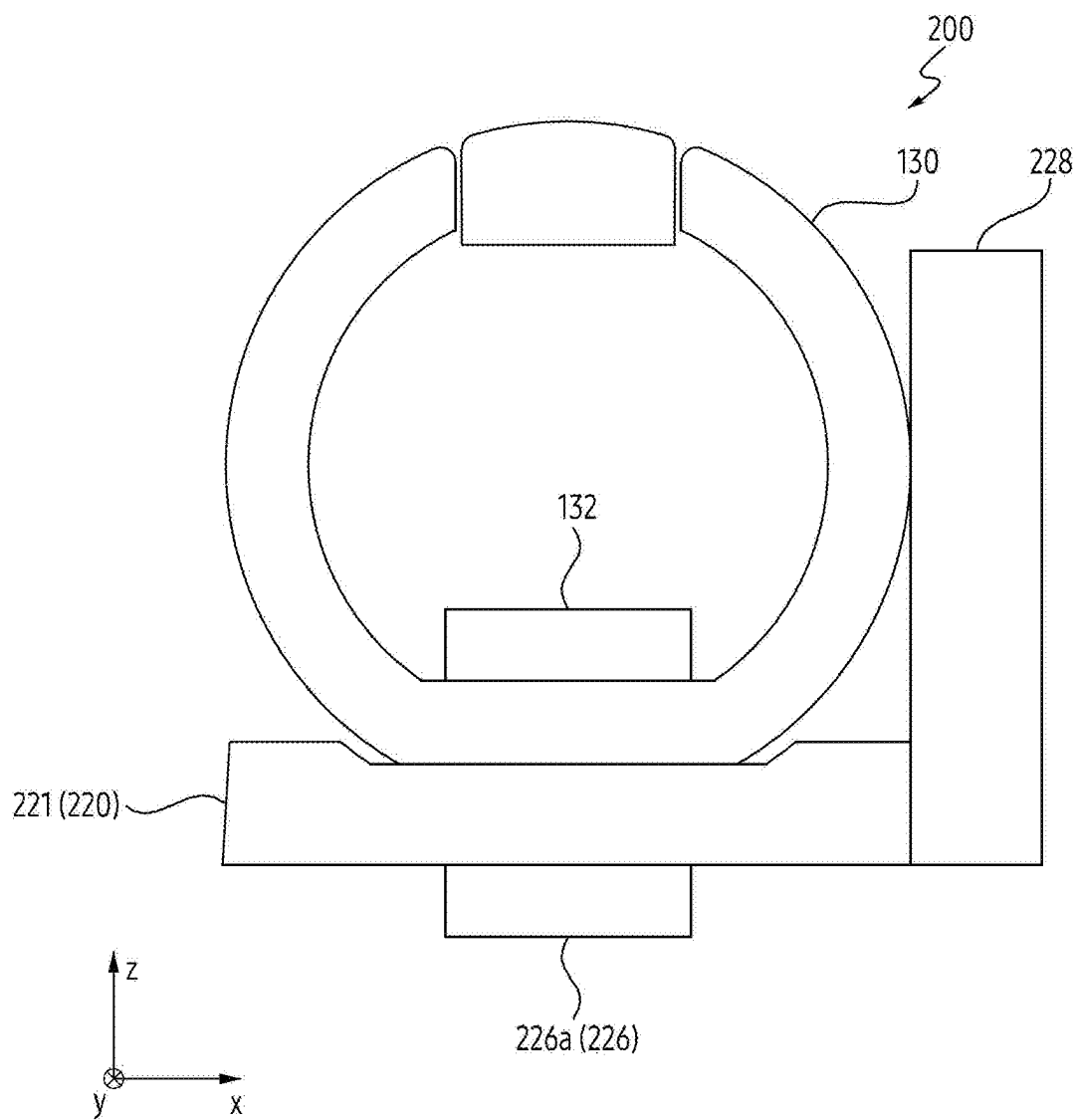


FIG. 4B

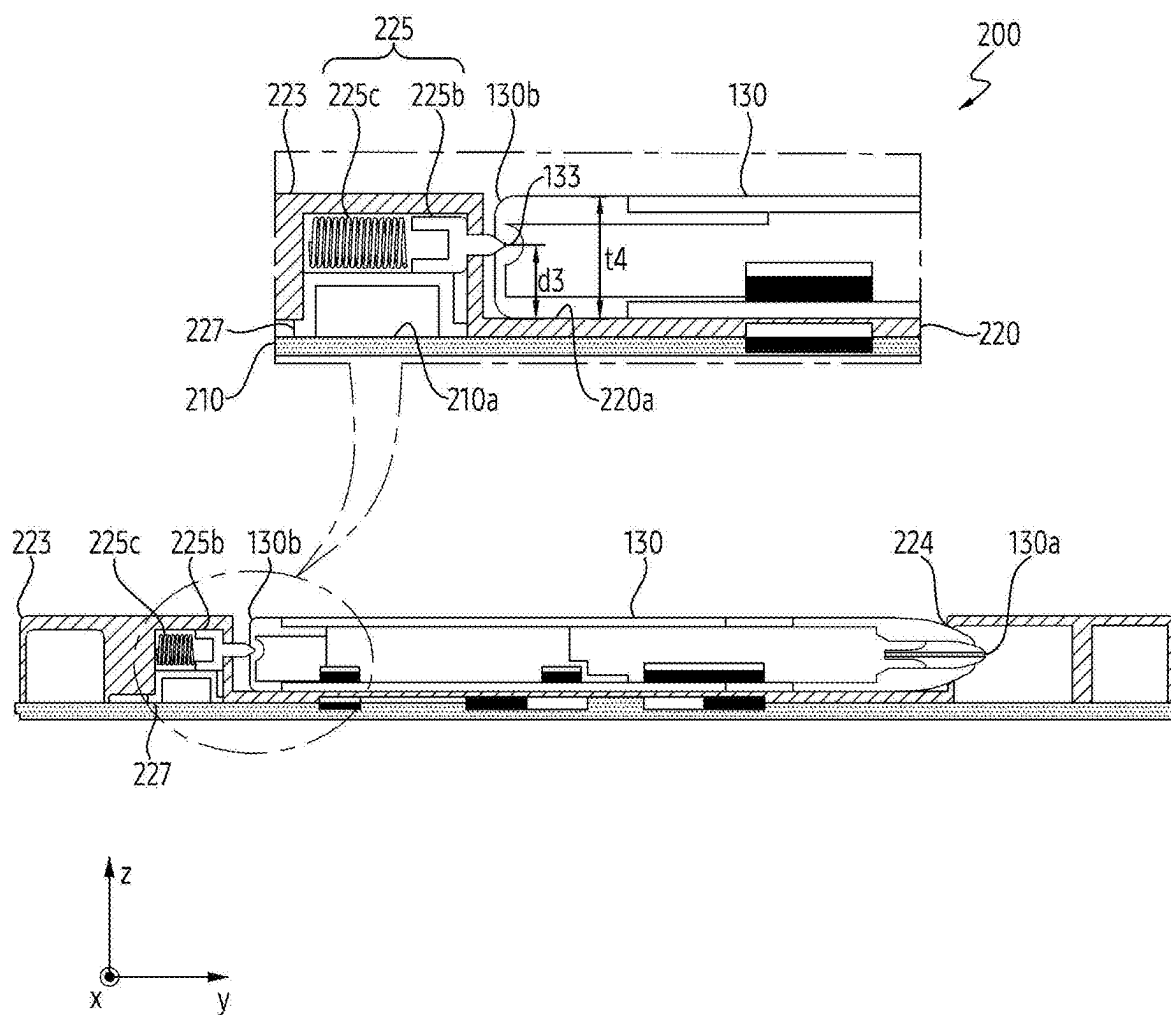


FIG. 5

CASE INCLUDING STRUCTURE FOR RECEIVING ELECTRONIC PEN

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application is a continuation application, claiming priority under 35 U.S.C. § 365 (c), of an International application No. PCT/KR2023/017114, filed on Oct. 31, 2023, which is based on and claims the benefit of a Korean patent application number 10-2023-0004889, filed on Jan. 12, 2023, in the Korean Intellectual Property Office, and of a Korean patent application number 10-2023-0015506, filed on Feb. 6, 2023, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

[0002] The disclosure relates to a case including a structure for receiving an electronic pen.

2. Description of Related Art

[0003] An electronic device may be exposed to various environments, by being carried by a user. A case may protect the electronic device exposed to various environments, by being detachably coupled to the electronic device. For example, the case may protect the electronic device from an impact transmitted to the electronic device from the outside of the electronic device. In order to meet a demand of the user, the case may have various functions in addition to a function for protecting the electronic device.

[0004] The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

SUMMARY

[0005] Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide a case for including structure for receiving electronic pen.

[0006] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

[0007] In accordance with an aspect of the disclosure, a case detachably coupled to an electronic device is provided. The case includes a plate supporting the electronic device, a storing member for storing at least a portion of an electronic pen disposed on a surface of the plate, wherein the storing member includes a receiving space receiving an end of the electronic pen, a first actuating portion applying a force in a direction toward the receiving space to another end of the electronic pen opposite to the end of the electronic pen such that the end of the electronic pen is inserted into the receiving space, and a second actuating portion, disposed between the receiving space and the first actuating portion, and applying a force in a direction toward the surface of the plate to the electronic pen such that the electronic pen is seated onto the storing member, and wherein a portion of the

storing member opens toward an outside of the storing member such that at least a portion of the electronic pen is exposed outside the storing member.

[0008] In accordance with another aspect of the disclosure, a case detachably coupled to an electronic device is provided. The case includes a plate supporting the electronic device, and a storing member for storing at least a portion of an electronic pen disposed on a surface of the plate, wherein the storing member includes a seating portion, having a first thickness, on which the electronic pen is seated a receiving space receiving an end of the electronic pen, a first protruding portion, having a second thickness greater than the first thickness, surrounding the receiving space, and connected to an end of the seating portion, a first actuating portion applying a force in a direction toward the receiving space to another end of the electronic pen opposite to the end of the electronic pen such that the end of the electronic pen is inserted into the receiving space, a second protruding portion, having a third thickness greater than the first thickness, surrounding the first actuating portion, and connected to another end of the seating portion opposite to the end of the seating portion, and a second actuating portion, disposed in the seating portion, applying a force in a direction toward the surface of the plate to the electronic pen such that the electronic pen is seated onto the storing member.

[0009] Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0011] FIG. 1A is a perspective view illustrating an example in which a case and an electronic device are coupled according to an embodiment of the disclosure;

[0012] FIG. 1B is a top plan view illustrating an example in which a case and an electronic device are coupled according to an embodiment of the disclosure;

[0013] FIG. 1C is an exploded perspective view of a case according to an embodiment of the disclosure;

[0014] FIG. 2 is a cross-sectional view illustrating an example of a case cut along line A-A' of FIG. 1B according to an embodiment of the disclosure;

[0015] FIG. 3A is a perspective view of a case according to an embodiment of the disclosure;

[0016] FIG. 3B is a cross-sectional view illustrating an example of a case cut along line B-B' of FIG. 1B according to an embodiment of the disclosure;

[0017] FIG. 4A is a top plan view of a case according to an embodiment of the disclosure;

[0018] FIG. 4B is a cross-sectional view illustrating an example of a case cut along line C-C' of FIG. 4A according to an embodiment of the disclosure; and

[0019] FIG. 5 is a cross-sectional view illustrating an example of a case according to an embodiment of the disclosure.

[0020] The same reference numerals are used to represent the same elements throughout the drawings.

DETAILED DESCRIPTION

[0021] The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

[0022] The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

[0023] It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

[0024] It should be appreciated that the blocks in each flowchart and combinations of the flowcharts may be performed by one or more computer programs which include instructions. The entirety of the one or more computer programs may be stored in a single memory device or the one or more computer programs may be divided with different portions stored in different multiple memory devices.

[0025] Any of the functions or operations described herein can be processed by one processor or a combination of processors. The one processor or the combination of processors is circuitry performing processing and includes circuitry like an application processor (AP, e.g. a central processing unit (CPU)), a communication processor (CP, e.g., a modem), a graphics processing unit (GPU), a neural processing unit (NPU) (e.g., an artificial intelligence (AI) chip), a wireless fidelity (Wi-Fi) chip, a Bluetooth® chip, a global positioning system (GPS) chip, a near field communication (NFC) chip, connectivity chips, a sensor controller, a touch controller, a finger-print sensor controller, a display driver integrated circuit (IC), an audio CODEC chip, a universal serial bus (USB) controller, a camera controller, an image processing IC, a microprocessor unit (MPU), a system on chip (SoC), an IC, or the like.

[0026] FIG. 1A is a perspective view illustrating an example in which a case and an electronic device are coupled according to an embodiment of the disclosure.

[0027] FIG. 1B is a top plan view illustrating an example in which a case and an electronic device are coupled according to an embodiment of the disclosure.

[0028] FIG. 1C is an exploded perspective view of a case according to an embodiment of the disclosure.

[0029] Referring to FIGS. 1A, 1B, and 1C, according to an embodiment, an electronic device 100 may be detachably coupled to a case 200. The electronic device 100 may be

disposed on the case 200. According to another embodiment, the electronic device 100 may include a housing 110 and a display 120.

[0030] According to yet another embodiment, the housing 110 may define at least a portion of an outer surface of the electronic device 100. For example, the housing 110 may define at least a portion of a rear surface and a side surface of the electronic device 100. The rear surface of the electronic device 100 may mean a surface opposite to a front surface of the electronic device 100 facing a user while the user is provided with visual information through the electronic device 100. The side surface of the electronic device 100 may mean a surface connecting the front surface of the electronic device 100 and the rear surface of the electronic device 100. The housing 110 may provide an internal space in which at least a portion of components of the electronic device 100 are disposed. For example, the housing 110 may receive various electronic components of the electronic device 100 to provide various functions to the user. According to yet another embodiment, the housing 110 may directly contact the case 200 in a state in which the electronic device 100 and the case 200 are coupled to each other. For example, the housing 110 may be disposed on the case 200 in the state in which the electronic device 100 and the case 200 are coupled to each other.

[0031] According to yet another embodiment, the display 120 may provide visual information to the user. The display 120 may be configured to emit light for providing visual content to the user. According to yet another embodiment, the display 120 may be disposed on the housing 110. The display 120 may be coupled to the housing 110. For example, the display 120 may define at least a portion of the front surface of the electronic device 100 by being disposed on the housing 110. For example, the display 120 may face the user while the user is provided with visual information through the electronic device 100.

[0032] According to yet another embodiment, an electronic pen 130 may be configured to transmit a signal corresponding to an input of the user to the electronic device 100. The electronic pen 130 may be configured to transmit a signal indicating an input of the user received on the display 120 to the electronic device 100. The electronic pen 130 may be configured to transmit a signal indicating an input of the user received above the display 120 to the electronic device 100. For example, the electronic pen 130 may be configured to transmit an electromagnetic signal corresponding to an input of the user to the electronic device 100. For example, the electronic pen 130 may include a coil (not illustrated) for transmitting a user input to the electronic device 100 through an electromagnetic signal. For example, the electronic device 100 may include an electromagnetic induction panel (not illustrated) for receiving a user input from the electronic pen 130 through an electromagnetic signal. For example, the electronic pen 130 may include an electro-magnetic resonance (EMR) method. For example, the electromagnetic induction panel of the electronic device 100 may transmit a first electromagnetic signal that causes electromagnetic resonance in the electronic pen 130 to the electronic pen 130. The electronic pen 130 may obtain a second electromagnetic signal by a current caused by electromagnetic induction in the electronic pen 130 based on receiving the first electromagnetic signal. The electronic pen 130 may transmit the second electromagnetic signal causing electromagnetic resonance in the electronic device 100 to

the electromagnetic induction panel of the electronic device 100. The electronic device 100 may obtain a third electromagnetic signal by a current caused by electromagnetic induction based on receiving the second electromagnetic signal through the electromagnetic induction panel. The electronic device 100 may identify a type of the user input from the electronic pen 130 based on identifying a phase of the third electromagnetic signal. For example, the electronic device 100 may identify the user input from the electronic pen 130 as a touch input on the display 120 based on identifying that the phase of the third electromagnetic signal is less than or equal to a first threshold value. For example, the electronic device 100 may identify the user input from the electronic pen 130 as a hovering input above the display 120 based on identifying that the phase of the third electromagnetic signal is greater than the first threshold value. The electronic device 100 may obtain information on a position of the electronic pen 130 based on the third electromagnetic signal. For example, the information on the position of the electronic pen 130 may include coordinates on the display 120 corresponding to the position of the electronic pen 130 above the display 120. However, an operation method of the electronic pen 130 is not limited thereto. For example, the electronic device 100 may identify the user input from the electronic pen 130 through a capacitive sensor in the display 120. For example, the electronic pen 130 may include an active electrostatic (AES) type, a pressure-sensitive type, or an infrared/ultrasonic type pen.

[0033] According to yet another embodiment, the electronic pen 130 may substantially have a shape of a pen. For example, the electronic pen 130 may be referred to as a stylus pen. For example, an end 130a of the electronic pen 130 may have a shape of a pen tip. For example, the electronic pen 130, and the end 130a of the electronic pen 130 may include a material having flexibility. For example, another end 130b of the electronic pen 130 may be opposite to the end 130a of the electronic pen 130. For example, a length of the electronic pen 130 may be defined as a distance between the end 130a of the electronic pen 130 and the other end 130b of the electronic pen 130. A length of a component may mean a distance in a direction (e.g., a +y direction or a -y direction) parallel to a direction (e.g., the -y direction) from the end 130a of the electronic pen 130 toward the other end 130b of the electronic pen 130, and the corresponding expression may be utilized substantially the same below unless otherwise stated.

[0034] According to yet another embodiment, the case 200 may protect the electronic device 100, by being detachably coupled to the electronic device 100. The case 200 may buffer an impact transmitted to the electronic device 100 from the outside of the electronic device 100. According to yet another embodiment, the case 200 may include a plate 210 and/or a storing member 220.

[0035] According to yet another embodiment, the plate 210 may define at least a portion of an outer surface of the case 200. The plate 210 may support the electronic device 100. The plate 210 may contact the electronic device 100. For example, a portion of a surface 210a of the plate 210 may face the electronic device 100. A portion of the surface 210a of the plate 210 may be in contact with the housing 110 of the electronic device 100. According to yet another embodiment, a size of the plate 210 may be larger than a size of the electronic device 100. The plate 210 may include a first portion 211, a second portion 212, and/or a third portion

213. According to yet another embodiment, the first portion 211 may support the electronic device 100. The first portion 211 may be in contact with the electronic device 100. The electronic device 100 may be disposed on the first portion 211. For example, the electronic device 100 may be disposed on an inner surface 211a (e.g., the first inner surface 211a) of the first portion 211. For example, the inner surface 211a of the first portion 211 may face the rear surface of the electronic device 100. The second portion 212 may be movable with respect to the first portion 211. For example, the second portion 212 may be rotatable with respect to the first portion 211. For example, the second portion 212 may be foldable with respect to the first portion 211. An inner surface 212a (e.g., the second inner surface 212a) of the second portion 212 may be spaced apart from the inner surface 211a (e.g., the first inner surface 211a) of the first portion 211. The third portion 213 may be disposed between the first portion 211 and the second portion 212. The third portion 213 may connect the first portion 211 and the second portion 212. For example, an end of the third portion 213 may be connected to the first portion 211, and another end of the third portion 213 may be connected to the second portion 212. An inner surface 213a (e.g., the third inner surface 213a) of the third portion 213 may be disposed between the inner surface 211a (e.g., the first inner surface 211a) of the first portion 211 and the inner surface 212a (e.g., the second inner surface 212a) of the second portion 212. The inner surface 213a of the third portion 213 may connect the inner surface 211a of the first portion 211 and the inner surface 212a of the second portion 212. The surface 210a of the plate 210 may include the inner surface 211a of the first portion 211, the inner surface 212a of the second portion 212, and the inner surface 213a of the third portion 213. The surface 210a of the plate 210 may be defined by the inner surface 211a of the first portion 211, the inner surface 212a of the second portion 212, and the inner surface 213a of the third portion 213.

[0036] According to yet another embodiment, as the first portion 211 and the second portion 212 are movable with respect to each other, a state of the case 200 may include a first state and a second state. In the first state of the case 200, a direction (e.g., a +z direction) in which the inner surface 211a of the first portion 211 faces may be substantially the same as a direction (e.g., the +z direction) in which the inner surface 212a of the second portion 212 faces. In the first state of the case 200, the display 120 may be visually exposed to the outside of the case 200. In the first state of the case 200, the display 120 may not be covered by the second portion 212. As the second portion 212 moves with respect to the first portion 211, the state of the case 200 may be changed from the first state to the second state. In the second state of the case 200, the inner surface 211a of the first portion 211 and the inner surface 212a of the second portion 212 may face each other. For example, in the second state of the case 200, the direction in which the inner surface 211a of the first portion 211 faces and the direction in which the inner surface 212a of the second portion 212 faces may be opposite to each other. In the second state of the case 200, the display 120 may be covered by the second portion 212. As the second portion 212 moves with respect to the first portion 211, the state of the case 200 may be changed from the second state to the first state.

[0037] According to yet another embodiment, the storing member 220 may accommodate at least a portion of the

electronic pen 130. The storing member 220 may support the electronic pen 130. The electronic pen 130 may be disposed on the storing member 220. For example, the electronic pen 130 may be disposed on a surface 220a of the storing member 220. According to yet another embodiment, the storing member 220 may be disposed on the plate 210. For example, the storing member 220 may be disposed on the surface 210a of the plate 210. The storing member 220 may be disposed between the first portion 211 and the second portion 212. The storing member 220 may be disposed on the third portion 213. For example, the storing member 220 may be disposed on the inner surface 213a of the third portion 213. According to yet another embodiment, the storing member 220 may include a seating portion 221, a first protruding portion 222, a second protruding portion 223, a receiving space 224, a first actuating portion 225, a second actuating portion 226, and/or a supporting member 227.

[0038] According to yet another embodiment, the seating portion 221 may support the electronic pen 130. The electronic pen 130 may be seated on the seating portion 221. The seating portion 221 may be disposed between the first protruding portion 222 and the second protruding portion 223. The seating portion 221 may define a step between the first protruding portion 222 and the second protruding portion 223. For example, the seating portion 221 may have a thickness thinner than a thickness of each of the first protruding portion 222 and the second protruding portion 223. A thickness of a component may mean a distance in a direction (e.g., the +z direction or a -z direction) parallel to a direction (e.g., the +z direction) in which the surface 210a of the plate 210 faces in the first state of the case 200, and the corresponding expression may be utilized substantially the same below unless otherwise stated. According to yet another embodiment, a length of the seating portion 221 may correspond to a length of the electronic pen 130. For example, the length of the seating portion 221 may be substantially the same as the length of the electronic pen 130, but is not limited thereto. For example, the length of the seating portion 221 may be longer or shorter than the length of the electronic pen 130.

[0039] According to yet another embodiment, the first protruding portion 222 may receive the end 130a of the electronic pen 130. For example, the first protruding portion 222 may surround the end 130a of the electronic pen 130. For example, the first protruding portion 222 may envelop the end 130a of the electronic pen 130. The first protruding portion 222 may form the receiving space 224. The first protruding portion 222 may surround the receiving space 224. The first protruding portion 222 may envelop the receiving space 224. The first protruding portion 222 may have a thickness thicker than a thickness of the seating portion 221. The first protruding portion 222 may be connected to the seating portion 221. For example, the first protruding portion 222 may be connected to an end of the seating portion 221.

[0040] According to yet another embodiment, the second protruding portion 223 may be connected to the seating portion 221. The second protruding portion 223 may be connected to another end of the seating portion 221 opposite to the end of the seating portion 221 connected to the first protruding portion 222. According to yet another embodiment, the second protruding portion 223 may receive the first actuating portion 225. For example, the second pro-

truding portion 223 may surround the first actuating portion 225. For example, the second protruding portion 223 may envelop the first actuating portion 225. The second protruding portion 223 may have a thickness thicker than the thickness of the seating portion 221. For example, the thickness of the second protruding portion 223 may be substantially the same as the thickness of the first protruding portion 222, but is not limited thereto. For example, the thickness of the second protruding portion 223 may be different from the thickness of the first protruding portion 222.

[0041] According to yet another embodiment, the receiving space 224 may receive a portion of the electronic pen 130. The receiving space 224 may receive the end 130a of the electronic pen 130. The electronic pen 130 may be inserted into the receiving space 224. In a state in which the end 130a of the electronic pen 130 is disposed in the receiving space 224, a portion of the electronic pen 130 may be surrounded (or enveloped) by the first protruding portion 222. For example, the receiving space 224 may be defined as an empty space disposed inside the first protruding portion 222.

[0042] According to yet another embodiment, the first actuating portion 225 may detachably couple the electronic pen 130 and the case 200. The first actuating portion 225 may insert a portion of the electronic pen 130 into the receiving space 224. The first actuating portion 225 may apply a force in a first direction (e.g., the +y direction) toward the receiving space 224 to the other end 130b of the electronic pen 130. The first actuating portion 225 may move the electronic pen 130 in the first direction (e.g., the +y direction). The first actuating portion 225 may reduce movement in a second direction (e.g., the -y direction) opposite to the first direction (e.g., the +y direction) of the electronic pen 130. For example, the first actuating portion 225 may apply the force in the first direction (e.g., the +y direction) from the end 130a of the electronic pen 130 toward the other end 130b of the electronic pen 130. For example, the first actuating portion 225 may be described as a repelling portion (or an repelling structure) that applies a force to push the other end 130b of the electronic pen 130 in a direction (e.g., the +y direction) away from the first actuating portion 225. For example, the first actuating portion 225 may be referred to as a first coupling portion that applies a force in the direction (e.g., the +y direction) away from the first actuating portion 225 to the other end 130b of the electronic pen 130. According to yet another embodiment, the first actuating portion 225 may be disposed in the storing member 220. The first actuating portion 225 may be disposed in the second protruding portion 223. The first actuating portion 225 may be spaced apart from the surface 210a of the plate 210 in the second protruding portion 223. For example, the first actuating portion 225 may be spaced apart from the inner surface 213a of the third portion 213 in the second protruding portion 223.

[0043] According to yet another embodiment, the second actuating portion 226 may seat the electronic pen 130 on the storing member 220. The second actuating portion 226 may apply a force to the electronic pen 130 such that the electronic pen 130 maintains a state in which the electronic pen 130 is seated on the storing member 220. For example, the second actuating portion 226 may apply a force in a third direction (e.g., the -z direction) toward the surface 210a of

the plate 210 to the electronic pen 130. For example, the second actuating portion 226 may apply the force in the third direction (e.g., the $-z$ direction) approaching the surface 210a of the plate 210 to the electronic pen 130. For example, the second actuating portion 226 may apply the force in the third direction (e.g., the $-z$ direction) opposite to a direction (e.g., the $+z$ direction) in which the surface 210a of the plate 210 faces in the first state of the case 200 to the electronic pen 130. According to yet another embodiment, the second actuating portion 226 may be disposed in the storing member 220. The second actuating portion 226 may be surrounded by the seating portion 221 of the storing member 220. The second actuating portion 226 may be enveloped by the seating portion 221 of the storing member 220. The second actuating portion 226 may be attached on the surface 210a of the plate 210. For example, the second actuating portion 226 may be attached on the inner surface 213a of the third portion 213 in the seating portion 221. For example, the second actuating portion 226 may be received inside a groove formed by recessing the surface 210a of the plate 210 inwardly. For example, the second actuating portion 226 may be referred to as a second coupling portion that applies a force in a direction (e.g., the $-z$ direction) toward the surface 210a of the plate 210 to the electronic pen 130.

[0044] According to yet another embodiment, the supporting member 227 may space apart the first actuating portion 225 from the surface 210a of the plate 210. For example, the first actuating portion 225 may be spaced apart from the inner surface 213a of the third portion 213 by the supporting member 227. The supporting member 227 may be disposed between the surface 210a of the plate 210 and the first actuating portion 225. For example, the supporting member 227 may be in contact with the surface 210a of the plate 210 and the first actuating portion 225.

[0045] According to yet another embodiment, the electronic pen 130 may be easily coupled to the case 200 by the first actuating portion 225 and the second actuating portion 226. For example, a process in which the electronic pen 130 is coupled to the case 200 by the first actuating portion 225 and the second actuating portion 226 may be described with reference to FIG. 2.

[0046] FIG. 2 is a cross-sectional view illustrating an example of a case cut along line A-A' of FIG. 1B according to an embodiment of the disclosure.

[0047] Referring to FIG. 2, according to an embodiment, a first actuating portion 225 may include a first magnet 225a. The first magnet 225a may be disposed in a second protruding portion 223.

[0048] According to another embodiment, an electronic pen 130 may include a second magnet 131 corresponding to the first magnet 225a. The second magnet 131 may be disposed in the electronic pen 130. The second magnet 131 may be closer to another end 130b from among an end 130a and the other end 130b of the electronic pen 130. For example, the second magnet 131 may be disposed in the other end 130b of the electronic pen 130.

[0049] According to yet another embodiment, the first actuating portion 225 may insert the end 130a of the electronic pen 130 into a receiving space 224. The first magnet 225a of the first actuating portion 225 may insert the end 130a of the electronic pen 130 into the receiving space 224 by interacting with the second magnet 131. For example, the first magnet 225a may apply a force in a first direction (e.g., a $+y$ direction) to the other end 130b of the

electronic pen 130 by exerting a repulsive force to the second magnet 131. The electronic pen 130 may move along the first direction (e.g., the $+y$ direction) by receiving the force in the first direction (e.g., the $+y$ direction) from the first magnet 225a. As the electronic pen 130 moves in the first direction (e.g., the $+y$ direction), the end 130a of the electronic pen 130 may be inserted into the receiving space 224. The first actuating portion 225 may provide convenience for a user to easily couple the electronic pen 130 to a case 200 by providing the force in the first direction to the electronic pen 130.

[0050] According to yet another embodiment, the first actuating portion 225 may maintain a state in which the electronic pen 130 is inserted into the receiving space 224. For example, in case that the electronic pen 130 moves in a second direction (e.g., a $-y$ direction) opposite the first direction (e.g., the $+y$ direction), the electronic pen 130 may be pulled out from the receiving space 224. When the electronic pen 130 is pulled out from the receiving space 224, the electronic pen 130 may be separated from the case 200 regardless of an intention of the user. When the electronic pen 130 is separated from the case 200 regardless of the intention of the user, the user may lose the electronic pen 130. The case 200 according to yet another embodiment may maintain a position of the electronic pen 130 in the case 200 by the first actuating portion 225 that applies the force in the first direction toward the receiving space 224 to the electronic pen 130. As the position of the electronic pen 130 in the case 200 is maintained, a possibility of the electronic pen 130 being lost may be reduced.

[0051] According to yet another embodiment, a first distance d1 between a surface 220a of a storing member 220 in contact with the electronic pen 130 and the first magnet 225a of the first actuating portion 225 may be different from a second distance d2 between the surface 220a of the storing member 220 and the second magnet 131 in the electronic pen 130. The surface 220a of the storing member 220 may face a fourth direction (e.g., a $+z$ direction) opposite to a third direction (e.g., a $-z$ direction) approaching a surface 210a of a plate 210. For example, the first distance d1 between the surface 220a of the storing member 220 and a first center c1 of the first magnet 225a may be different from the second distance d2 between the surface 220a of the storing member 220 and a second center c2 of the second magnet 131 in the electronic pen 130. For example, the first distance d1 between the surface 220a of the storing member 220 and the first center c1 of the first magnet 225a may be greater than or equal to the second distance d2 between the surface 220a of the storing member 220 and the second center c2 of the second magnet 131 in the electronic pen 130. The first center c1 may be positioned on a center point of a first side surface 225a-1 of the first magnet 225a facing the fourth direction (e.g., the $+z$ direction) and a second side surface 225a-2 of the first magnet 225a opposite to the first side surface 225a-1. The second side surface 225a-2 of the first magnet 225a may face the third direction (e.g., the $-z$ direction). The second center c2 may be positioned on a center point of a first side surface 131a of the second magnet 131 facing the fourth direction (e.g., the $+z$ direction) and a second side surface 131b of the second magnet 131 opposite to the first side surface 131a. The second side surface 131b of the second magnet 131 may face the third direction (e.g., the $-z$ direction). However, it is not limited thereto. For example, the first distance d1 may mean a distance between

the surface **220a** of the storing member **220** and the side surface **225a-1** of the first magnet **225a** facing the fourth direction (e.g., the +z direction), and the second distance d2 may mean a distance between the surface **220a** of the storing member **220** and the side surface **131a** of the second magnet **131** in the electronic pen **130** facing the fourth direction (e.g., the +z direction). For example, the first distance d1 between the surface **220a** of the storing member **220** and the side surface **225a-1** of the first magnet **225a** facing the fourth direction (e.g., the +z direction) may be greater than or equal to the second distance d2 between the surface **220a** of the storing member **220** and the side surface **131a** of the second magnet **131** in the electronic pen **130** facing the fourth direction. For example, as the first magnet **225a** is supported by a supporting member **227** disposed in the second protruding portion **223**, the first distance d1 may be greater than or equal to the second distance d2. For example, in case that the second distance d2 is smaller than the first distance d1, the electronic pen **130** may receive a force in a direction inclined counterclockwise with respect to the first direction (e.g., the +y direction) by the first actuating portion **225** while being seated on the storing member **220**. In case that the electronic pen **130** receives the force in the direction inclined counterclockwise with respect to the first direction (e.g., the +y direction), the end **130a** of the electronic pen **130** may not move into the receiving space **224**. Since the first distance d1 has a size greater than or equal to the second distance d2, the case **200** according to yet another embodiment may provide a structure in which the end **130a** of the electronic pen **130** is easily inserted into the receiving space **224**. For example, in case that the first distance d1 is greater than the second distance d2, the first actuating portion **225** may provide the electronic pen **130** with a force inclined clockwise with respect to the first direction (e.g., the +y direction) while the electronic pen **130** is seated on the storing member **220**. As the electronic pen **130** is provided with the force inclined clockwise with respect to the first direction (e.g., +y direction), the end **130a** of the electronic pen **130** may be easily inserted into the receiving space **224**.

[0052] According to yet another embodiment, a second actuating portion **226** may include at least one third magnet **226a**. The at least one third magnet **226a** may be disposed in a seating portion **221**. The at least one third magnet **226a** may be disposed on the surface **210a** of the plate **210**. The at least one third magnet **226a** may be disposed in the seating portion **221**. The at least one third magnet **226a** may be disposed between the first actuating portion **225** and the receiving space **224**. The at least one third magnet **226a** may be disposed between the end **130a** of the electronic pen **130** and the other end **130b** of the electronic pen **130** in a state in which the electronic pen **130** is seated on the storing member **220**. The at least one third magnet **226a** may include a plurality of third magnets spaced apart from each other.

[0053] According to yet another embodiment, the electronic pen **130** may include at least one fourth magnet **132** corresponding to the at least one third magnet **226a**. The at least one fourth magnet **132** may be disposed in the electronic pen **130**. For example, the at least one fourth magnet **132** may be disposed between the end **130a** of the electronic pen **130** and the other end **130b** of the electronic pen **130**. According to yet another embodiment, the number of the at least one fourth magnet **132** may be substantially the same as the number of the at least one third magnet **226a**. For

example, the at least one fourth magnet **132** may include a plurality of fourth magnets spaced apart from each other. However, it is not limited thereto. For example, the number of the at least one fourth magnet **132** may be different from the number of the at least one third magnet **226a**.

[0054] According to yet another embodiment, the second actuating portion **226** may detachably attach the electronic pen **130** on the surface **220a** of the storing member **220**. For example, the at least one third magnet **226a** of the second actuating portion **226** may attach the electronic pen **130** on the surface **220a** of the storing member **220** by interacting with the at least one fourth magnet **132** in the electronic pen **130**. For example, the at least one third magnet **226a** may apply a force in the third direction (e.g., the -z direction) to the electronic pen **130** by exerting an attractive force to the at least one fourth magnet **132** disposed in the electronic pen **130**. As the force in the third direction (e.g., the -z direction) is applied to the electronic pen **130**, the electronic pen **130** may maintain a state of being attached on the surface **220a** of the storing member **220**.

[0055] According to yet another embodiment, magnitude of a force applied between the first magnet **225a** and the second magnet **131** may be greater than magnitude of a force applied between the at least one third magnet **226a** and the at least one fourth magnet **132**. As the magnitude of the force applied between the first magnet **225a** and the second magnet **131** is relatively greater, when the electronic pen **130** is inserted into the receiving space **224**, the electronic pen **130** may move along the first direction (e.g., the +y direction) while the at least one third magnet **226a** and the at least one fourth magnet **132** interact.

[0056] According to yet another embodiment, the seating portion **221**, which is a portion of the storing member **220**, may have a shape that is open toward the outside of the storing member **220**. For example, the seating portion **221** may be opened toward the outside of the storing member **220** by having a thickness thinner than a first protruding portion **222** and the second protruding portion **223**. The seating portion **221** may have a first thickness t1. The first protruding portion **222** may have a second thickness t2 greater than the first thickness t1. The second protruding portion **223** may have a third thickness t3 greater than the first thickness t1. As the seating portion **221** has the open shape, at least a portion of the electronic pen **130** may be exposed to the outside of the storing member **220** in a state of being seated in the storing member **220**. For example, at least a portion of the electronic pen **130** may not be covered by the storing member **220** in the state of being seated in the storing member **220**. According to yet another embodiment, the second thickness t2 and the third thickness t3 may be substantially the same. For example, the second thickness t2 may be substantially the same as a sum of the first thickness t1 and a fourth thickness t4, which is a thickness of the electronic pen **130**. For example, the third thickness t3 may be substantially the same as the sum of the first thickness t1 and the fourth thickness t4, which is the thickness of the electronic pen **130**. However, it is not limited thereto. For example, the second thickness t2 and the third thickness t3 may be different from each other.

[0057] As described above, the case **200** according to yet another embodiment may provide a structure in which the electronic pen **130** may be easily inserted into the receiving space **224** by the first actuating portion **225** that transmits the

force in the first direction (e.g., the +y direction) to the other end **130b** of the electronic pen **130**.

[0058] FIG. 3A is a perspective view of a case according to an embodiment of the disclosure.

[0059] FIG. 3B is a cross-sectional view illustrating an example of a case cut along line B-B' of FIG. 1B according to an embodiment of the disclosure.

[0060] Referring to FIGS. 3A and 3B, in a first state of a case **200**, a first portion **211** may form a step with respect to a third portion **213**. In the first state of the case **200**, a step may be formed at a boundary between the first portion **211** and the third portion **213**. In the first state of the case **200**, a second portion **212** may form a step with respect to the third portion **213**. In the first state of the case **200**, a step may be formed at a boundary between the second portion **212** and the third portion **213**.

[0061] According to an embodiment, a cross-sectional area of a receiving space **224** may be smaller than a cross-sectional area of an electronic pen **130**. For example, a width of the receiving space **224** may be smaller than a width of the electronic pen **130**. A width of a component may mean a distance in a direction (e.g., a +x direction or a -x direction) perpendicular to a first direction (e.g., a +y direction) and a third direction (e.g., a -z direction), and the corresponding expression may be utilized substantially the same below unless otherwise stated. As the cross-sectional area of the receiving space **224** is smaller than the cross-sectional area of the electronic pen **130**, a movement range of the electronic pen **130** inserted into the receiving space **224** by a first actuating portion (e.g., the first actuating portion **225** of FIG. 1C) may be reduced (or limited). As the movement range of the electronic pen **130** inserted into the receiving space **224** is reduced (or limited), at least a portion of the electronic pen **130** may be exposed to the outside of a storing member **220** such that a user may easily grip the electronic pen **130**.

[0062] According to another embodiment, the receiving space **224** may include a first region **224a** and a second region **224b**. The first region **224a** may receive a portion of the electronic pen **130**. The portion of the electronic pen **130** may be inserted into the first region **224a**. The first region **224a** may have a shape corresponding to a shape of the electronic pen **130**. For example, in case that a cross section of the electronic pen **130** has a substantially circular shape, a cross section of the first region **224a** may have a substantially circular shape. The second region **224b** may be connected to the first region **224a**. A size of the second region **224b** may be smaller than a size of the first region **224a**. For example, a width of the second region **224b** may be smaller than a width of the first region **224a**. As the size of the second region **224b** is smaller than the size of the first region **224a**, movement of the electronic pen **130** in the third direction (e.g., the -z direction) and/or a fourth direction (e.g., a +z direction) may be limited.

[0063] According to yet another embodiment, a surface **210a** of a plate **210** may be closer to the second region **224b** from among the first region **224a** and the second region **224b**. For example, a distance from the surface **210a** of the plate **210** to the first region **224a** may be greater than a distance from the surface **210a** of the plate **210** to the second region **224b**.

[0064] Referring to FIG. 3B, according to an embodiment, a seating portion **221** may include a protrusion **221a**. The protrusion **221a** may reduce a movement range of the

electronic pen **130** with respect to the storing member **220**. For example, the protrusion **221a** may reduce movement of the electronic pen **130** to the outside of the storing member **220**. The protrusion **221a** may protrude from a surface **220a** of the storing member **220** on which the electronic pen **130** is disposed. For example, the protrusion **221a** may extend along the fourth direction (e.g., the +z direction) from the surface **220a** of the storing member **220**. At least a portion of the protrusion **221a** may be in contact with at least a portion of the electronic pen **130** disposed on the surface **220a** of the storing member **220**. For example, at least a portion of the protrusion **221a** may be in contact with a periphery of at least a portion of the electronic pen **130**.

[0065] Referring back to FIG. 3A, according to an embodiment, a first protruding portion **222** may include a friction member **222a**. The friction member **222a** may limit the movement of the electronic pen **130** to the outside of the receiving space **224**. The friction member **222a** may be in contact with a portion of the electronic pen **130** inserted into the receiving space **224**. For example, the friction member **222a** may be in contact with an outer surface of the portion of the electronic pen **130** inserted into the receiving space **224**. As the electronic pen **130** is provided with a friction force due to contact with the friction member **222a**, movement of the electronic pen **130** may be limited. For example, the friction member **222a** may be manufactured from a material (e.g., thermoplastic polyurethane (TPU), and/or silicone) for providing a friction force to the electronic pen **130**. The friction member **222a** may be disposed inside the receiving space **224**. For example, the friction member **222a** may be disposed along an inner surface of the receiving space **224**. For example, the friction member **222a** may be disposed on an inner surface of the first protruding portion **222** along a periphery of the receiving space **224**.

[0066] As described above, since the size of the second region **224b** is smaller than the size of the first region **224a** of the receiving space **224**, the case **200** according to another embodiment may provide a structure in which a position of the electronic pen **130** inserted into the receiving space **224** may be maintained. The case **200** according to yet another embodiment may provide a structure in which a position of the electronic pen **130** seated on the storing member **220** may be maintained by the protrusion **221a** and the friction member **222a** that reduce the movement range of the electronic pen **130**.

[0067] FIG. 4A is a top plan view of a case according to an embodiment of the disclosure.

[0068] FIG. 4B is a cross-sectional view illustrating an example of a case cut along line C-C' of FIG. 4A according to an embodiment of the disclosure.

[0069] Referring to FIGS. 4A and 4B, according to an embodiment, a storing member **220** may include a barrier **228**. The barrier **228** may limit movement of an electronic pen **130** with respect to the storing member **220**. The barrier **228** may reduce a movement range of the electronic pen **130** with respect to the storing member **220**. According to another embodiment, the barrier **228** may be disposed on a surface **210a** of a plate **210**. The barrier **228** may be connected to a seating portion **221**. The barrier **228** may be connected to at least one of a first protruding portion **222** and a second protruding portion **223**. For example, the barrier **228** may be connected to the first protruding portion **222**. For example, the barrier **228** may be connected to the second protruding portion **223**. For example, the barrier **228** may be

connected to the first protruding portion 222 and the second protruding portion 223. A length of the barrier 228 may be less than or equal to a length of the electronic pen 130. For example, the length of the barrier 228 may be shorter than a length of the seating portion 221. For example, the length of the barrier 228 may be substantially the same as the length of the seating portion 221.

[0070] According to yet another embodiment, the barrier 228 may shield a magnetic field from a third magnet 226a of a second actuating portion 226 and/or a fourth magnet 132 in the electronic pen 130. For example, the barrier 228 may include a conductive material (e.g., a metallic material). As the barrier 228 shields the magnetic field from at least one of the third magnet 226a and the fourth magnet 132, malfunction of an electronic device 100 disposed on a case 200 may be reduced.

[0071] According to yet another embodiment, a thickness of the barrier 228 may be thicker than a thickness of the seating portion 221. For example, the thickness of the barrier 228 may be substantially the same as a sum of the thickness of the seating portion 221 and a thickness of the electronic pen 130. For example, the thickness of the barrier 228 may be smaller than the sum of the thickness of the seating portion 221 and the thickness of the electronic pen 130. As the barrier 228 is disposed, movement of the electronic pen 130 to the outside of the storing member 220 may be reduced (or limited).

[0072] Meanwhile, in FIGS. 4A and 4B, the barrier 228 is illustrated as being connected to a side surface of the seating portion 221 facing a +x-axis direction, but is not limited thereto. For example, the barrier 228 may be connected to another side surface of the seating portion 221 facing a -x-axis direction. For example, the barrier 228 may be connected to the side surface of the seating portion 221 facing the +x-axis direction and the other side surface of the seating portion 221 facing the -x-axis direction.

[0073] As described above, the case 200 according to yet another embodiment may provide a structure in which a position of the electronic pen 130 may be maintained by the barrier 228 that limits the movement range of the electronic pen 130 with respect to the storing member 220.

[0074] FIG. 5 is a cross-sectional view illustrating an example of a case according to an embodiment of the disclosure.

[0075] Referring to FIG. 5, according to an embodiment, a first actuating portion 225 may include a pressing member 225b (or a pressing portion 225b) and an elastic member 225c. The first pressing portion 225b may be movable along a first direction (e.g., a +y direction) or a second direction (e.g., a -y direction) opposite to the first direction with respect to a storing member 220. For example, the first pressing portion 225b may move along the first direction (e.g., the +y direction) such that an end 130a of an electronic pen 130 is inserted into a receiving space 224. According to an embodiment, at least a portion of the first pressing portion 225b may be disposed in the storing member 220. The first pressing portion 225b may protrude from the inside of the storing member 220 to the outside of the storing member 220 by moving in the first direction (e.g., the +y direction). The first pressing portion 225b may be in contact with another end 130b of the electronic pen 130 by protruding to the outside of the storing member 220. The first pressing portion 225b may apply a force in the first direction (e.g., the

+y direction) to the electronic pen 130 by being in contact with the other end 130b of the electronic pen 130.

[0076] According to another embodiment, the elastic member 225c may provide a force to the pressing member 225b. The elastic member 225c may depress the pressing member 225b. For example, the elastic member 225c may provide a force in the first direction (e.g., the +y direction) to the pressing member 225b by being in contact with the pressing member 225b. For example, the elastic member 225c may provide an elastic force to the pressing member 225b by elastically deforming. For example, the elastic member 225c may be a spring, but is not limited thereto. For example, the elastic member 225c may be replaced with various members that follow Hooke's Law. According to yet another embodiment, the elastic member 225c may be disposed in the storing member 220. The elastic member 225c may be disposed in a second protruding portion 223 of the storing member 220.

[0077] According to yet another embodiment, a supporting member 227 may support the pressing member 225b of the first actuating portion 225. For example, the supporting member 227 may be disposed between the pressing member 225b and a surface 210a of a plate 210. The supporting member 227 may support the elastic member 225c of the first actuating portion 225. For example, the supporting member 227 may be disposed between the elastic member 225c and the surface 210a of the plate 210.

[0078] According to yet another embodiment, the electronic pen 130 may include a receiving groove 133. The receiving groove 133 may be disposed in the other end 130b of the electronic pen 130. For example, as a portion of the other end 130b of the electronic pen 130 is recessed inwardly into the electronic pen 130, the receiving groove 133 may be formed. The receiving groove 133 may receive a portion of the pressing member 225b. For example, the receiving groove 133 may have a shape corresponding to a portion of the pressing member 225b. For example, while the pressing member 225b protrudes outside the second protruding portion 223, the electronic pen 130 may approach the storing member 220. As the electronic pen 130 approaches the storing member 220, the other end 130b of the electronic pen 130 may contact the pressing member 225b. The pressing member 225b may move inside the storing member 220 along the second direction (e.g., the -y direction) by contact with the other end 130b of the electronic pen 130. As the pressing member 225b moves in the second direction (e.g., the -y direction), the elastic member 225c may be compressed. While the pressing member 225b is inserted into the receiving groove 133 of the electronic pen 130, the compressed elastic member 225c may depress the pressing member 225b in the first direction (e.g., the +y direction). The pressing member 225b may move the electronic pen 130 in the first direction (e.g., the +y direction) based on receiving a force from the elastic member 225c. The electronic pen 130 may be inserted into the receiving space 224 by moving along the first direction (e.g., the +y direction) by the first pressing portion 225b. According to yet another embodiment, the pressing member 225b may apply a force in the first direction (e.g., the +y direction) to the other end 130b of the electronic pen 130 while the electronic pen 130 is inserted into the receiving space 224. As the pressing member 225b provides the force in the first direction (e.g., the +y direction) to the electronic pen 130, a

position of the electronic pen **130** may maintain a state of being inserted into the receiving space **224**.

[0079] According to yet another embodiment, a third distance **d3** between a surface **220a** of the storing member **220** on which the electronic pen **130** is disposed and the pressing member **225b** may be different from half of a fourth thickness **t4**, which is a thickness of the electronic pen **130**. For example, the third distance **d3** may be greater than or equal to half of the fourth thickness **t4**. In case that the third distance **d3** is less than half of the fourth thickness **t4**, the electronic pen **130** may receive a force inclined counterclockwise with respect to the first direction (e.g., the +y direction) while being seated on the storing member **220**. In case that the electronic pen **130** receives the force inclined counterclockwise with respect to the first direction (e.g., the +y direction), the end **130a** of the electronic pen **130** may not move into the receiving space **224**. Since the third distance **d3** is greater than or equal to half of the fourth thickness **t4**, a case **200** according to yet another embodiment may provide a structure in which the end **130a** of the electronic pen **130** is easily inserted into the receiving space **224**.

[0080] As described above, the case **200** according to yet another embodiment may provide a structure in which the electronic pen **130** is easily seated on the storing member **220** by the first actuating portion **225** including the pressing member **225b** and the elastic member **225c**.

[0081] A case may include a structure for receiving an electronic pen for transmitting a user input to an electronic device. When the user uses the electronic pen, the electronic pen may be separated from the case. The user may detachably couple the electronic pen to the case after using the electronic pen. When the user couples the electronic pen to the case, a structure for easily receiving the electronic pen in the case may be required. When the user carries the case in a state of receiving the electronic pen in the case, a structure in which the electronic pen is not easily separated from the case may be required.

[0082] According to yet another embodiment, a case (e.g., the case **200** of FIGS. **1A**, **1B**, and **1C**) detachably coupled to an electronic device (e.g., the electronic device **100** of FIGS. **1A** and **1B**) is provided. According to yet another embodiment, the case may comprise a plate (e.g., the plate **210** of FIGS. **1A**, **1B**, and **1C**) supporting the electronic device. According to yet another embodiment, the case may comprise a storing member (e.g., the storing member **220** of FIGS. **1A**, **1B**, and **1C**) for storing at least a portion of an electronic pen (e.g., the electronic pen **130** of FIGS. **1A**, **1B**, and **1C**) disposed on a surface (e.g., the surface **210a** of FIGS. **1A**, **1B**, and **1C**) of the plate. According to yet another embodiment, the storing member may include a receiving space (e.g., the receiving space **224** of FIGS. **1A** and **1C**) receiving an end (e.g., the end **130a** of FIGS. **1A** and **1C**) of the electronic pen. According to yet another embodiment, the storing member may include a first actuating portion (e.g., the first actuating portion **225** of FIG. **1C**) applying a force in a direction toward the receiving space to another end (e.g., the other end **130b** of FIGS. **1A** and **1C**) of the electronic pen opposite to the end of the electronic pen such that the end of the electronic pen is inserted into the receiving space. According to yet another embodiment, the storing member may include a second actuating portion (e.g., the second actuating portion **226** of FIG. **1C**), disposed between the receiving space and the first actuating portion, and applying a force in a direction toward the surface of the

plate to the electronic pen such that the electronic pen is seated onto the storing member. According to yet another embodiment, a portion of the storing member may open toward an outside of the storing member such that at least a portion of the electronic pen is exposed outside the storing member.

[0083] The case according to yet another embodiment may provide a structure in which the electronic pen may be easily inserted into the receiving space by the first actuating portion that transmits a force in a direction toward the receiving space to the other end of the electronic pen. The electronic pen according to yet another embodiment may provide a structure in which the electronic pen is easily coupled to the case by the second actuating portion that provides a force for seating the electronic pen on the storing member. Since the storing member is open toward the outside, the electronic device according to yet another embodiment may provide a structure in which a user may easily separate the electronic pen coupled to the case.

[0084] According to yet another embodiment, the first actuating portion may include a first magnet (e.g., the first magnet **225a** of FIG. **2**). According to yet another embodiment, the first magnet may exert a repulsive force to a second magnet (e.g., the second magnet **131** of FIG. **2**) disposed inside the another end of the electronic pen, thereby inserting the end of the electronic pen into the receiving space.

[0085] The case according to yet another embodiment may provide a structure in which the electronic pen may be easily inserted into the receiving space by the first magnet that transmits a force in a direction toward the receiving space to the other end of the electronic pen.

[0086] According to yet another embodiment, a distance (e.g., the first distance **d1** of FIG. **2**) between a surface of the storing member on which the electronic pen is disposed and a center (e.g., the first center **c1** of FIG. **2**) of the first magnet may be greater than or equal to a distance (e.g., the second distance **d2** of FIG. **2**) between the surface of the storing member and a center (e.g., the second center **c2** of FIG. **2**) of the second magnet.

[0087] According to yet another embodiment, a distance (e.g., the first distance **d1** of FIG. **2**) between a side surface (e.g., the side surface **225a-1** of FIG. **2**) of the first magnet facing a direction in which a surface (e.g., the surface **220a** of FIG. **2**) of the storing member on which the electronic pen is disposed faces and the surface of the storing member may be greater than or equal to a distance (e.g., the second distance **d2** of FIG. **2**) between a side surface (e.g., the side surface **131a** of FIG. **2**) of the second magnet facing a direction in which the surface of the storing member faces and the surface of the storing member.

[0088] Since the first distance between the storing member and the first magnet is greater than or equal to a distance between the storing member and the second magnet, the case according to yet another embodiment may provide a structure in which the electronic pen may be easily inserted into the receiving space.

[0089] According to yet another embodiment, the storing member may include a supporting member (e.g., the supporting member **227** of FIG. **2**) supporting the first magnet by being disposed between the first magnet and the surface of the plate.

[0090] Since the first distance between the storing member and the first magnet is greater than or equal to the distance

between the storing member and the second magnet, by the supporting member supporting the first magnet, the case according to yet another embodiment may provide the structure in which the electronic pen may be easily inserted into the receiving space.

[0091] According to yet another embodiment, the first actuating portion may include a pressing member (e.g., the pressing member **225b** of FIG. 5) that moves in a first direction such that the another end of the electronic pen is inserted into the receiving space or in a second direction opposite to the first direction. According to yet another embodiment, the first actuating portion may include an elastic member (e.g., the pressing member **225b** of FIG. 5), disposed in the storing member, depressing the pressing member.

[0092] The case according to yet another embodiment may provide a structure in which the electronic pen may be easily inserted into the receiving space by the pressing member that transmits a force in a direction toward the receiving space to the other end of the electronic pen.

[0093] According to yet another embodiment, a distance (e.g., the third distance d_3 of FIG. 5) between a surface of the storing member on which the electronic pen is disposed and the pressing member may be greater than or equal to half a thickness (e.g., the fourth thickness t_4 of FIG. 5) of the electronic pen.

[0094] Since the third distance between the storing member and the pressing member is greater than or equal to half of the thickness of the electronic pen, the case according to yet another embodiment may provide the structure in which the electronic pen may be easily inserted into the receiving space.

[0095] According to yet another embodiment, the storing member may include a supporting member (e.g., the supporting member **227** of FIG. 5) disposed between the surface of the plate and the pressing member.

[0096] Since the third distance between the storing member and the pressing member is greater than or equal to half of the thickness of the electronic pen, by the supporting member supporting the first magnet, the case according to yet another embodiment may provide the structure in which the electronic pen may be easily inserted into the receiving space.

[0097] According to yet another embodiment, the storing member may include a seating portion (e.g., the seating portion **221** of FIG. 1C) on which the electronic pen is seated. According to an embodiment, the storing member may include a first protruding portion (e.g., the first protruding portion **222** of FIG. 1C), connected to an end of the seating portion, surrounding the receiving space, and having a greater thickness than the seating portion. According to yet another embodiment, the storing member may include a second protruding portion (e.g., the second protruding portion **223** of FIG. 1C), connected to another end of the seating portion opposite to the end of the seating portion, surrounding the first actuating portion, and having a thickness greater than a thickness of the seating portion.

[0098] The case according to yet another embodiment may provide a structure in which a portion of the storing member is opened toward the outside by the first protruding portion and the second protruding portion defining a step with respect to the seating portion.

[0099] According to yet another embodiment, the seating portion may include a protrusion (e.g., the protrusion **221a**

of FIG. 3B), protruding from a surface of the storing member on which the electronic pen is disposed, and surrounding the electronic pen.

[0100] The case according to yet another embodiment may provide a structure capable of reducing a movement range of the electronic pen with respect to the storing member by the protrusion formed on the seating portion.

[0101] According to yet another embodiment, the storing member may include a barrier (e.g., the barrier **228** of FIGS. 4A and 4B), having a thickness greater than the thickness of the seating portion, and connected to the second protruding portion.

[0102] The case according to yet another embodiment may provide the structure capable of reducing the movement range of the electronic pen with respect to the storing member by the barrier connected to the first protruding portion.

[0103] According to yet another embodiment, the receiving space may include a first region (e.g., the first region **224a** of FIG. 3A) having a shape corresponding to a shape of the electronic pen. According to yet another embodiment, the receiving space may include a second region (e.g., the second region **224b** of FIG. 3A), having a width smaller than a width of the first region, connected to the first region. According to yet another embodiment, the surface of the plate may be closer to the second region from among the first region and the second region.

[0104] Since the second region close to the plate is smaller than the first region far from the plate, the case according to yet another embodiment may provide the structure capable of reducing the movement range of the electronic pen.

[0105] According to yet another embodiment, the plate may include a first portion (e.g., the first portion **211** of FIG. 1C) on which the electronic device is disposed. According to yet another embodiment, the plate may include a second portion (e.g., the second portion **212** of FIG. 1C) being foldable with respect to the first portion. According to yet another embodiment, the plate may include a third portion (e.g., the third portion **213** of FIG. 1C) disposed between the first portion and the second portion. According to yet another embodiment, the storing member may be disposed on the third portion.

[0106] The case according to yet another embodiment may provide a structure capable of covering the electronic device by the second portion movable with respect to the first portion.

[0107] According to yet another embodiment, a state of the case may include a first state in which a direction in which an inner surface (e.g., the inner surface **211a** of FIG. 1C) of the first portion facing the electronic device faces and a direction in which an inner surface (e.g., the inner surface **212a** of FIG. 1C) of the second portion faces are the same, and a second state in which the inner surface of the first portion and the inner surface of the second portion face each other. According to yet another embodiment, in the first state, a step may be formed at a boundary between the first portion and the third portion.

[0108] The case according to yet another embodiment may provide the structure capable of covering the electronic device by the second portion movable with respect to the first portion.

[0109] According to yet another embodiment, the second actuating portion may include at least one third magnet (e.g., the at least one third magnet **226a** of FIGS. 3A and 3B)

disposed between the first actuating portion and the receiving space. According to yet another embodiment, the at least one third magnet may exert an attractive force to at least one fourth magnet (e.g., the at least one fourth magnet **132** of FIGS. 3A and 3B) that corresponds to the at least one third magnet and is disposed in the electronic pen.

[0110] The case according to yet another embodiment may provide a structure capable of maintaining a position of the electronic pen on the storing member by the at least one third magnet disposed between the end of the electronic pen and the other end of the electronic pen.

[0111] According to yet another embodiment, the case may comprise a friction member (e.g., the friction member **222a** of FIG. 3A), disposed along an inner surface of the receiving space, in contact with a portion of the electronic pen inserted into the receiving space.

[0112] The case according to yet another embodiment may provide the structure capable of reducing the movement range of the electronic pen with respect to the receiving space by the friction member disposed in the receiving space.

[0113] According to yet another embodiment, a case (e.g., the case **200** of FIGS. 1A, 1B, and 1C) detachably coupled to an electronic device (e.g., the electronic device **100** of FIGS. 1A and 1B) is provided. According to yet another embodiment, the case may comprise a plate (e.g., the plate **210** of FIGS. 1A, 1B, and 1C) supporting the electronic device. According to yet another embodiment, the case may comprise a storing member (e.g., the storing member **220** of FIGS. 1A, 1B, and 1C) for storing at least a portion of an electronic pen (e.g., the electronic pen **130** of FIGS. 1A, 1B, and 1C) disposed on a surface (e.g., the surface **210a** of FIGS. 1A, 1B, and 1C) of the plate. According to yet another embodiment, the storing member may include a seating portion (e.g., the seating portion **221** of FIG. 1C), having a first thickness (e.g., the first thickness t_1 of FIG. 2), on which the electronic pen is seated. According to yet another embodiment, the storing member may include a receiving space (e.g., the receiving space **224** of FIG. 1C) receiving an end (e.g., the end **130a** of FIGS. 1A and 1C) of the electronic pen. According to yet another embodiment, the storing member may include a first protruding portion (e.g., the first protruding portion **222** of FIG. 1C), having a second thickness (e.g., the second thickness t_2 of FIG. 2) greater than the first thickness, surrounding the receiving space, and connected to an end of the seating portion. According to yet another embodiment, the storing member may include a first actuating portion (e.g., the first actuating portion **225** of FIG. 1C) applying a force in a direction toward the receiving space to another end of the electronic pen opposite to the end of the electronic pen such that the end of the electronic pen is inserted into the receiving space. According to yet another embodiment, the storing member may include a second protruding portion (e.g., the second protruding portion **223** of FIG. 1C), having a third thickness (e.g., the third thickness t_3 of FIG. 2) greater than the first thickness, surrounding the first actuating portion, and connected to another end (e.g., the other end **130b** of FIGS. 1A and 1C) of the seating portion opposite to the end of the seating portion. According to yet another embodiment, the storing member may include a second actuating portion (e.g., the second actuating portion **226** of FIG. 1C), disposed in the seating portion, applying a

force in a direction toward the surface of the plate to the electronic pen such that the electronic pen is seated onto the storing member.

[0114] The case according to yet another embodiment may provide a structure in which the electronic pen may be easily inserted into the receiving space by the first actuating portion that transmits a force in a direction toward the receiving space to the other end of the electronic pen. The electronic pen according to yet another embodiment may provide a structure in which the electronic pen is easily coupled to the case by the second actuating portion that provides a force for seating the electronic pen on the storing member. Since the storing member is open toward the outside, the electronic device according to yet another embodiment may provide a structure in which a user may easily separate the electronic pen coupled to the case.

[0115] According to yet another embodiment, the first actuating portion may include a first magnet (e.g., the first magnet **225a** of FIG. 2). According to yet another embodiment, the first magnet may exert a repulsive force to a second magnet (e.g., the second magnet **131** of FIG. 2) disposed inside the another end of the electronic pen, thereby inserting the end of the electronic pen into the receiving space.

[0116] The case according to yet another embodiment may provide a structure in which the electronic pen may be easily inserted into the receiving space by the first magnet that transmits a force in a direction toward the receiving space to the other end of the electronic pen.

[0117] According to yet another embodiment, the first actuating portion may include a pressing member (e.g., the pressing member **225b** of FIG. 5) that moves in a first direction such that the another end of the electronic pen is inserted into the receiving space or in a second direction opposite to the first direction. According to yet another embodiment, the first actuating portion may include an elastic member (e.g., the pressing member **225b** of FIG. 5), disposed in the storing member, and depressing the pressing member.

[0118] The case according to yet another embodiment may provide a structure in which the electronic pen may be easily inserted into the receiving space by the pressing member that transmits a force in a direction toward the receiving space to the other end of the electronic pen.

[0119] According to yet another embodiment, the seating portion may include a protrusion (e.g., the protrusion **221a** of FIG. 3B), protruding from a surface of the storing member on which the electronic pen is disposed, and surrounding the electronic pen.

[0120] The case according to yet another embodiment may provide a structure capable of reducing a movement range of the electronic pen with respect to the storing member by the protrusion formed on the seating portion.

[0121] According to yet another embodiment, the receiving space may include a first region (e.g., the first region **224a** of FIG. 3A) having a shape corresponding to a shape of the electronic pen. According to yet another embodiment, the receiving space may include a second region (e.g., the second region **224b** of FIG. 3A), having a width smaller than a width of the first region, connected to the first region. According to yet another embodiment, the surface of the plate may be closer to the second region from among the first region and the second region.

[0122] Since the second region close to the plate is smaller than the first region far from the plate, the case according to yet another embodiment may provide the structure capable of reducing the movement range of the electronic pen.

[0123] The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to yet another embodiment of the disclosure, the electronic devices are not limited to those described above.

[0124] It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” or “connected with” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0125] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0126] While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

What is claimed is:

1. A case detachably coupled to an electronic device, the case comprising:

a plate supporting the electronic device; and
a storing member for storing at least a portion of an electronic pen disposed on a surface of the plate, wherein the storing member includes:

a receiving space receiving an end of the electronic pen,
a first actuating portion applying a force in a direction toward the receiving space to another end of the electronic pen opposite to the end of the electronic pen such that the end of the electronic pen is inserted into the receiving space, and
a second actuating portion, disposed between the receiving space and the first actuating portion, and applying a force in a direction toward the surface of the plate to the electronic pen such that the electronic pen is seated onto the storing member, and

wherein a portion of the storing member is open toward an outside of the storing member such that at least a portion of the electronic pen is exposed outside the storing member.

2. The case of claim 1,

wherein the first actuating portion includes a first magnet, and

wherein the first magnet exerts a repulsive force to a second magnet disposed inside the another end of the electronic pen, thereby inserting the end of the electronic pen into the receiving space.

3. The case of claim 2, wherein a distance between a surface of the storing member on which the electronic pen is disposed and a center of the first magnet is greater than or equal to a distance between the surface of the storing member and a center of the second magnet.

4. The case of claim 2, wherein the storing member further includes a supporting member supporting the first magnet by being disposed between the first magnet and the surface of the plate.

5. The case of claim 1, wherein the first actuating portion includes:

a pressing member that moves in a first direction such that the another end of the electronic pen is inserted into the receiving space or in a second direction opposite to the first direction; and

an elastic member, disposed in the storing member, depressing the pressing member.

6. The case of claim 5, wherein a distance between a surface of the storing member on which the electronic pen is disposed and the pressing member is greater than or equal to half a thickness of the electronic pen.

7. The case of claim 5, wherein the storing member further includes a supporting member disposed between the surface of the plate and the pressing member.

8. The case of claim 1, wherein the storing member includes:

a seating portion on which the electronic pen is seated;
a first protruding portion, connected to an end of the seating portion, surrounding the receiving space, and having a greater thickness than the seating portion; and
a second protruding portion, connected to another end of the seating portion opposite to the end of seating portion, surrounding the first actuating portion, and having a thickness greater than a thickness of the seating portion.

9. The case of claim 8, wherein the seating portion includes a protrusion, protruding from a surface of the storing member on which the electronic pen is disposed, and surrounding the electronic pen.

10. The case of claim 8, wherein the storing member includes a barrier, having a thickness greater than the thickness of the seating portion, and connected to the second protruding portion.

11. The case of claim 1,

wherein the receiving space includes:

a first region having a shape corresponding to a shape of the electronic pen, and

a second region, having a width smaller than a width of the first region, connected to the first region, and wherein the surface of the plate is closer to the second region from among the first region and the second region.

12. The case of claim 1,

wherein the plate includes:

a first portion on which the electronic device is disposed,

a second portion being foldable with respect to the first portion, and

a third portion disposed between the first portion and the second portion, and

wherein the storing member is disposed on the third portion.

13. The case of claim 12,

wherein a state of the case includes:

a first state in which a direction in which an inner surface of the first portion facing the electronic device faces and a direction in which an inner surface of the second portion faces are the same, and

a second state in which the inner surface of the first portion and the inner surface of the second portion face each other, and

wherein, in the first state, a step is formed at a boundary between the first portion and the third portion.

14. The case of claim 1,

wherein the second actuating portion includes at least one third magnet disposed between the first actuating portion and the receiving space, and

wherein the at least one third magnet exerts an attractive force to at least one fourth magnet that corresponds to the at least one third magnet and is disposed in the electronic pen.

15. The case of claim 1, further comprising:

a friction member, disposed along an inner surface of the receiving space, in contact with a portion of the electronic pen inserted into the receiving space.

16. A case detachably coupled to an electronic device, the case comprising:

a plate supporting the electronic device; and

a storing member for storing at least a portion of an electronic pen disposed on a surface of the plate,

wherein the storing member includes:

a seating portion, having a first thickness, on which the electronic pen is seated,

a receiving space receiving an end of the electronic pen, a first protruding portion, having a second thickness greater than the first thickness, surrounding the receiving space, and connected to an end of the seating portion,

a first actuating portion applying a force in a direction toward the receiving space to another end of the electronic pen opposite to the end of the electronic pen such that the end of the electronic pen is inserted into the receiving space,

a second protruding portion, having a third thickness greater than the first thickness, surrounding the first actuating portion, and connected to another end of the seating portion opposite to the end of the seating portion, and

a second actuating portion, disposed in the seating portion, applying a force in a direction toward the surface of the plate to the electronic pen such that the electronic pen is seated onto the storing member.

17. The case of claim 16,

wherein the first actuating portion includes a first magnet, and

wherein the first magnet exerts a repulsive force to a second magnet disposed inside the another end of the electronic pen, thereby inserting the end of the electronic pen into the receiving space.

18. The case of claim 16, wherein the first actuating portion includes:

a pressing member that moves in a first direction such that the another end of the electronic pen is inserted into the receiving space or in a second direction opposite to the first direction; and

an elastic member, disposed in the storing member, and depressing the pressing member.

19. The case of claim 16, wherein the seating portion includes a protrusion, protruding from a surface of the storing member on which the electronic pen is disposed, and surrounding the electronic pen.

20. The case of claim 16,

wherein the receiving space includes:

a first region having a shape corresponding to a shape of the electronic pen, and

a second region, having a width smaller than a width of the first region, connected to the first region, and

wherein the surface of the plate is closer to the second region than the first region.

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